

Improved Opacification of a Suboptimally Enhanced Pulmonary Artery in Chest CT: Experience Using a Dual-Layer Detector Spectral CT

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OBJECTIVE. The objective of our study was to evaluate the quality of virtual monoenergetic imaging (VMI) from dual-layer detector spectral CT and the effect of virtual monoenergetic images obtained at low energies on the detection of pulmonary embolism (PE) in patients with a suboptimally enhanced pulmonary artery on chest CT.

MATERIALS AND METHODS. Of 1552 consecutive chest CT examinations performed with dual-layer detector spectral CT using a routine protocol with a tube voltage of 120 kVp, 79 examinations with suboptimal enhancement of the pulmonary artery (i.e., mean attenuation of pulmonary artery ≤ 180 HU) were included. The mean attenuation of the pulmonary artery, noise, contrast-to-noise ratio (CNR), and signal-to-noise ratio (SNR) of virtual monoenergetic images obtained at 40–200 keV were compared with those of the conventional 120-kVp images. The virtual monoenergetic images with the best CNR were compared with the 120-kVp images with regard to subjective image quality and diagnostic accuracy for detecting PE.

RESULTS. Sufficient attenuation of the pulmonary artery (> 180 HU) was obtained using VMI for 78 of the 79 examinations. The noise levels of the virtual monoenergetic images were gradually increased with decreasing energy level (i.e., kiloelectron volt setting). The CNR and SNR of virtual monoenergetic images at 40–65 keV were significantly higher (both, $p < 0.001$) than the CNR and SNR of the 120-kVp images. The CNR was the highest at 40 keV for all cases. Diagnostic accuracy for detecting PE was significantly higher for 40-keV images (reader 1: AUC = 0.992, $p = 0.033$; reader 2: AUC = 0.986, $p = 0.043$) than for 120-kVp images (reader 1, AUC = 0.911; reader 2, AUC = 0.933). The subjective quality was not different between these two images.

CONCLUSION. In chest CT examinations in which the pulmonary artery is suboptimally enhanced, obtaining virtual monoenergetic images at a low energy setting using dual-layer detector spectral CT allows sufficient attenuation of the pulmonary artery to be achieved while preserving image quality and increasing diagnostic performance for detecting PE.

Keywords: chest CT, pulmonary artery, spectral CT, suboptimal contrast enhancement, virtual monoenergetic image, virtual monoenergetic imaging (VMI)

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Routine chest CT examinations are performed for various clinical indications such as trauma, evaluation of cancer, and nonspecific chest pain. Contrast enhancement is useful in routine chest CT for characterization and differentiation of parenchymal lesions from surrounding normal structures and the assessment of various vascular conditions such as bleeding, hemoptysis, and aneurysm. Therefore, the goal of contrast enhancement in routine chest CT is to achieve a diagnostically adequate enhancement of thoracic cardiovascular structures [1]. Sufficient attenuation for the depiction of an endo- or perivascular abnormality is known to be greater than 180 HU for the pulmonary artery and greater than 200 HU for the aorta [2, 3]. However, the pul-

monary artery can show insufficient enhancement on chest CT because of flow-related artifact, insufficient amount of contrast administered, or improper scan timing, thus limiting the ability to diagnose unsuspected pulmonary embolism (PE) [2]. When chest CT shows equivocal findings for the presence of PE because of suboptimal enhancement, pulmonary CT angiography (CTA) would be necessary. This additional imaging study increases the burden of radiation exposure and IV contrast material to patients.

Virtual monoenergetic images extrapolated from dual-energy CT (DECT) data can improve vascular opacification at low energy levels. The performance of virtual monoenergetic imaging (VMI) for pulmonary CTA with a reduced iodine dose has been studied

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by various groups [3–5]. Recently released noise-optimized VMI techniques applied to CTA for imaging various body parts have shown improved image quality compared with traditional virtual monoenergetic images, particularly at 40 or 50 keV [6–8].

Dual-layer detector spectral CT has been recently introduced [9]. This system consists of a single tube and two-layered detectors where high- and low-energy photons are simultaneously captured during each CT examination. The top layer selectively absorbs low-energy photons while the bottom layer absorbs high-energy photons. Because energy separation is done on the detector, the decision to use a single- or dual-energy protocol before scanning is not necessary, and spectral information can be obtained from all examinations performed using a routine 120-kVp protocol [9, 10].

We hypothesized that the generation of spectral data using the same protocol with MDCT might facilitate the practical use of spectral information and improve image quality and diagnostic performance in suboptimal studies. Therefore, the purpose of this study was to evaluate image quality of virtual monoenergetic images obtained from dual-layer detector spectral CT and the effect of these images on the detection of incidental PE compared with a conventional 120-kVp image in patients with suboptimal enhancement of the pulmonary artery on routine chest CT.

Materials and Methods

Patient Selection

This study was approved by the institutional review board. Written informed consent was waived because of its retrospective nature. Between August 2016 and March 2017, 1552 consecutive adult patients (≥ 18 years) underwent contrast-enhanced chest CT using dual-layer detector spectral CT. The clinical indications for the examinations were chest pain, dyspnea on exertion, general weakness, hemoptysis, trauma, and cancer evaluation. CT scans obtained using a pulmonary angiography protocol with clinical suspicion of PE were not included. One radiologist (with 18 years of experience in chest CT) reviewed all chest CT scans on a PACS (Universal Viewer 6.0, GE Healthcare) and measured the attenuation of the pulmonary artery at the trunk and both main arteries using circular ROIs at each site. Two patients were excluded from analysis because of severe motion artifact and beam-hardening artifact at the level of pulmonary artery, respectively.

Among the remaining 1550 patients, 79 patients had CT examinations with a mean attenuation of the pulmonary artery, which was obtained from three

measurements, of 180 HU or less and thus were considered to have CT examinations with insufficient pulmonary artery opacification [1,2] and were included in the study group. No patient had a patent foramen ovale, intracardiac shunt, or vascular malformation. Of these 79 patients, 53, 16, and 10 patients were referred from emergency department, inpatient clinic, and outpatient clinic, respectively. The reasons for suboptimal opacification of the pulmonary artery included transient contrast interruption ($n = 63$), delayed start of scanning due to concurrent examinations of another body part ($n = 6$), contrast extravasation ($n = 4$), extremely large body size of the patient ($n = 3$), erroneous ROI placement in bolus tracking ($n = 2$), and technical error ($n = 1$). Transient interruption of contrast material was considered to be the cause when there was poor opacification of the pulmonary artery even though the aorta was adequately enhanced and contrast material was still arriving from the superior vena cava. Patient characteristics and dose-length product data are summarized in Table 1.

Image Acquisition

All chest CT examinations were performed using a dual-layer detector spectral CT unit (IQon, Philips Healthcare). The scanning parameters were as follows: 120 kVp, 140–250 mA (reference mAs, 73), pitch of 0.609, rotation time of 0.4 second, 64×0.625 mm collimation, 2-mm slice thickness, 2-mm slice increment, and smooth filter (filter A). Scanning was performed in the cranio-caudal direction from the top to the bottom of the chest during a single deep-inspiration breath-hold. A total of 90 mL of iodinated contrast medium (iohexol [Omnipaque 300, GE Healthcare]) was injected via a peripheral vein at a rate of 2–2.5 mL/s followed by 50 mL of saline injected at the same rate. Individual contrast injection timing was controlled using bolus tracking in the aortic arch with the threshold set at 150 HU followed by a 7-second delay before scanning. After scanning, data from two detectors were combined in the projection domain. They were reconstructed with a standard kernel using an iterative reconstruction method (iDose2, Philips Healthcare) to obtain conventional 120-kVp images and were then transferred to PACS. Subsequently, a spectral CT-based image was reconstructed and transferred to a dedicated workstation. A spectral CT-based image is a DICOM file that includes all the data needed to generate spectral results. Using spectral CT-based imaging, virtual monoenergetic images were generated in the workstation integrated into our PACS.

Quantitative Image Analysis

One radiologist (with 5 years of experience in body imaging) measured the attenuation (mean

CT number in Hounsfield units) of the pulmonary artery and subscapularis muscle and measured the background noise (SD of the CT number in Hounsfield units) from conventional 120-kVp images on the PACS. The attenuation of the pulmonary artery was measured in the same way as was used in the patient selection process. The subscapularis muscle was measured for sections without beam hardening. Background noise was measured at room air 2 cm apart from the anterior chest wall on the midline. The size of each ROI was adapted to the diameter of the structure analyzed. All measurements were performed three times, and the mean of the three measurements was recorded as the representative value.

When a user clicked “Extend Applications” on an image, a dedicated workstation (Spectral Diagnostic Suite, Philips Healthcare) integrated into our PACS was opened and displayed on the same

TABLE 1: Characteristics of 79 Patients With Suboptimal Opacification of the Pulmonary Artery on Routine Chest CT and Dose-Length Product of CT Examinations

Patient or Examination Characteristic	Value
Sex, no. of patients	
Male	62
Female	17
Age (y)	
Mean	56
Range	19–93
Height (cm), mean $\pm$ SD	166.3 $\pm$ 9.3
Body weight (kg), mean $\pm$ SD	74 $\pm$ 18.4
Referral department, no. of patients	
Emergency department	53
Inpatient clinic	16
Outpatient clinic	10
Reason for suboptimal opacification of pulmonary artery, no. of patients	
Transient contrast interruption	63
Delayed start of scanning	6
Contrast extravasation	4
Extremely large body size	3
Erroneous ROI placement in bolus tracking	2
Technical error	1
Dose-length product (mGy $\times$ cm)	
Mean	327
Range	194–452

monitor. Virtual monoenergetic images were generated by selecting the “mono E” button on the image to be measured. The energy level was changed by moving the scroll bar under the image. All parameters measured on the conventional 120-kVp image were measured on the virtual monoenergetic images with an increment of 5 keV from 40 to 70 keV and with an increment of 10 keV from 70 to 200 keV. The ROI of each measurement could be automatically transferred to entire energy levels of the virtual monoenergetic images. The contrast-to-noise ratio (CNR) was calculated according to the following equation:

$$CNR = (SI_{PA} - SI_{muscle}) / BN,$$

where SI_{PA} is the mean signal intensity of the pulmonary artery, SI_{muscle} is the mean signal intensity of the subscapularis muscle, and BN is the background noise. In addition, the signal-to-noise ratio (SNR) was calculated according to the following equation:

$$SNR = SI_{PA} / BN,$$

where SI_{PA} is the mean signal intensity of the pulmonary artery, and BN is the background noise.

Qualitative Image Analysis

The subjective image quality of the virtual monoenergetic images with the best CNR and the subjective image quality of the conventional 120-kVp image with regard to overall noise and artifacts were assessed by two radiologists (with 5 and 19 years of experience in chest CT, respectively) in consensus using a 4-point scale (score = 0, complete or almost complete absence of either noise or artifacts; score = 1, mild noise or artifacts; score = 2, moderate noise or artifacts; score = 3, marked noise or artifacts). Readers were blinded to the image reconstruction technique. All images were analyzed with constant windowing (window width, 400 HU; window level, 60 HU).

Among 79 patients, PE was confirmed in six patients through pulmonary CTA performed within 24 hours ($n = 4$) or 48 hours ($n = 2$) after the chest CT examination with suboptimal enhancement. For these six cases, a consensus reading was performed by two radiologists (with 18 and 10 years of experience in chest CT, respectively) to provide a reference standard using all available images. Two radiologists (reader 1 with 6 years and reader 2 with 9 years of experience in vascular imaging) independently reviewed the conventional 120-kVp image and the virtual monoenergetic image with the best CNR in the

six patients with PE with a 2-week interval between reading sessions. At each reading session, the detection of PE was evaluated for the following eight anatomic segments of the pulmonary artery: pulmonary trunk, right main, left main, right upper lobe, right middle lobe, right lower lobe, left upper lobe, and left lower lobe. For each segment, the presence of PE was rated on a 5-point scale (1 = PE certainly absent, 2 = PE possibly absent, 3 = PE suspected, 4 = PE possibly present, 5 = PE definitely present).

Statistical Analysis

Results are expressed as means $\pm$ SD for continuous variables and frequencies or percentages for categorical variables. Comparisons between virtual monoenergetic images and the conventional 120-kVp image were performed using a paired t test for quantitative variables. Differences in subjective image quality between the conventional 120-kVp image and the virtual monoenergetic image with the best CNR were calculated using the Wilcoxon signed-rank test. The assessment of diagnostic accuracy in the detection of PE was conducted by calculating AUCs and corresponding 95% CIs from an ROC curve analysis for the virtual monoenergetic image with the best CNR and the 120-kVp image. All statistical analyses were per-

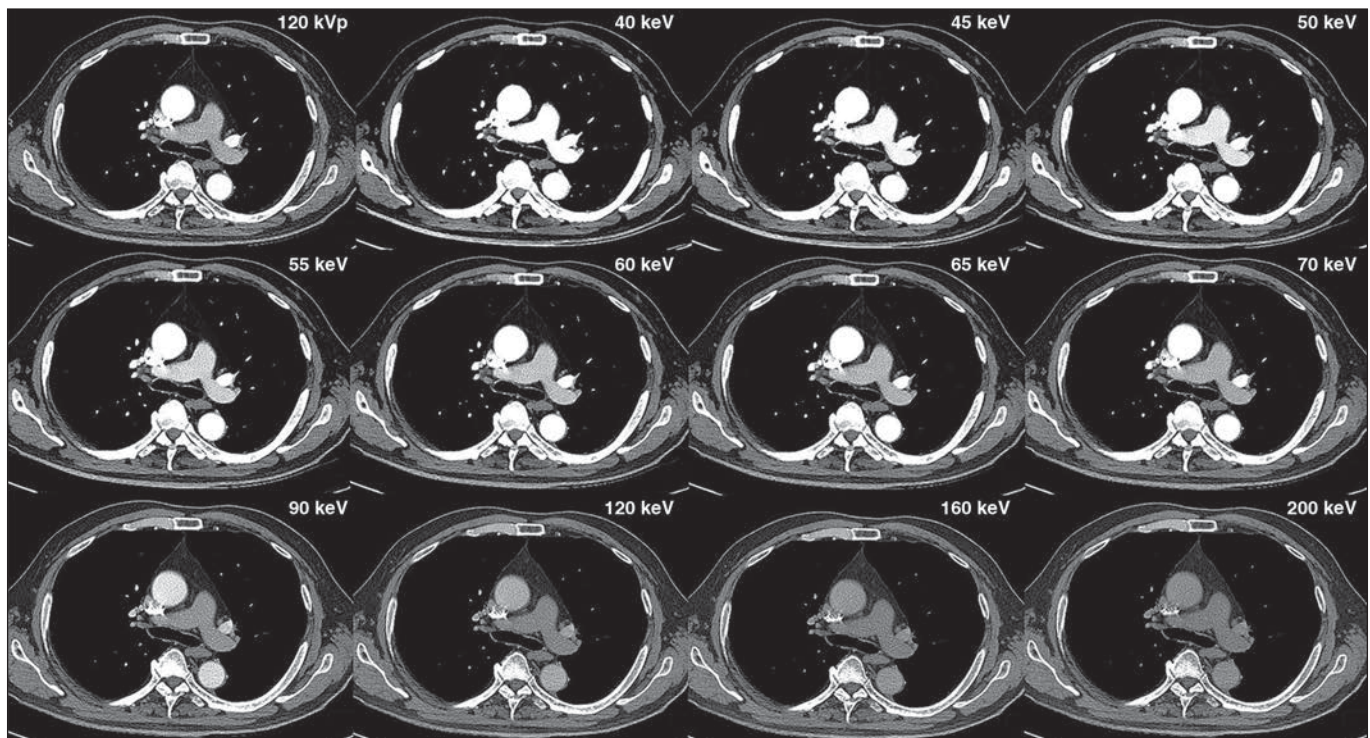


Fig. 1—69-year-old man with colon cancer. Image obtained from routine chest CT examination (top row, left image) for evaluation of metastasis shows pulmonary artery is not sufficiently opacified (116 HU) because of flow-related artifact. Note adequately enhanced aorta and contrast agent still arriving from superior vena cava. Virtual monoenergetic image reconstruction was performed, and virtual monoenergetic images (all other images) were obtained with energies ranging from 40 to 200 keV. Sufficient opacification of pulmonary artery (> 180 HU) was obtained at 40–50 keV.

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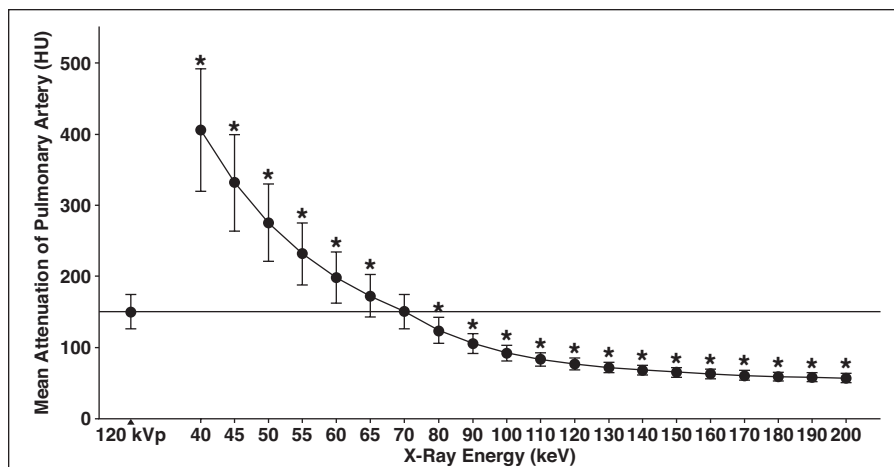


Fig. 2—Graph of mean vascular attenuation of pulmonary artery measured at each energy level in 79 patients with suboptimally enhanced pulmonary artery on chest CT. Horizontal line shows that mean vascular attenuation on conventional 120-kVp image is similar to that on 70-keV image. Mean vascular attenuation measured from virtual monoenergetic images increases with decreasing energy level. Compared with mean vascular attenuation of pulmonary artery at 120 kVp, pulmonary artery showed significantly higher attenuation at 40–65 keV but lower attenuation at 80–200 keV (both, $p < 0.001$). For each energy level, asterisk indicates statistically significant difference compared with 120 kVp. ● = mean, whiskers = ± 1 SD.

formed using commercially available software (MedCalc, version 17.6, MedCalc Software). A p value of < 0.05 was considered statistically significant. Interreader agreement was calculated with weighted kappa coefficients and were interpreted as follows: a kappa value of 0.20 or less, poor agreement; 0.21–0.40, fair agreement; 0.41–0.60, moderate agreement; 0.61–0.80, substantial agreement; and 0.81 or greater, almost perfect agreement.

Results

For 78 of 79 (98.7%) patients with suboptimal opacification of the pulmonary artery on the CT examination performed at 120 kVp, sufficient opacification (> 180 HU) was achieved at 40–70 keV. Sufficient attenuation of the pulmonary artery was obtained in one (1.3%) patient at 70 keV, 41 (51.9%) patients at 65 keV, 60 (75.9%) patients at 60 keV, 68 (86.1%) patients at 55 keV, 72 (91.1%) patients at 50 keV, 75 (94.9%) patients at 45 keV, and

78 (98.7%) patients at 40 keV. For one patient, the attenuation of the pulmonary artery was 79.9 HU on the 120-kVp image and did not exceed 180 HU on the virtual monoenergetic images at any energy level including 40 keV.

Quantitative Image Analysis

The mean attenuation of the pulmonary artery on the conventional 120-kVp image of the 79 patients was 150.2 ± 23.9 HU (range, 79.9–179.0 HU). In all patients, the mean attenuation of the pulmonary artery measured from the virtual monoenergetic images was increased with decreasing energy level (Fig. 1). The mean attenuation of the pulmonary artery on the virtual monoenergetic images at 40–65 keV was significantly ($p < 0.001$) higher than that on the conventional 120-kVp image. However, the mean attenuation of the pulmonary artery at 70 keV was not significantly different from that at 120 kVp, and the mean attenuation of the pulmonary

artery at 80–200 keV was significantly ($p < 0.001$) lower than that at 120 kVp (Fig. 2).

The mean background noises at 70 keV (13.8 ± 1.6 HU) and 65 keV (14.5 ± 1.7 HU) were not significantly different from that at 120 kVp (14.1 ± 1.6 HU). The mean background noise in the range of 40–70 keV gradually increased from 13.8 ± 1.6 HU to 24.1 ± 4.8 HU as the energy level was decreased. The mean background noise was nearly flat in the range of 80–200 keV. The mean background noise at 40–60 keV was significantly ($p < 0.05$) higher than that at 120 kVp, whereas the mean background noise at 80–200 keV was significantly ($p < 0.05$) lower than that at 120 kVp (Fig. 3).

The mean CNR at 70 keV (6.7 ± 1.9) was not significantly different from that at 120 kVp (6.9 ± 2.0). However, the mean CNR at 40–65 keV (8.1 ± 2.3 – 14.9 ± 4.3) was significantly ($p < 0.001$) higher than that at 120 kVp, and the mean CNR at 80–200 keV

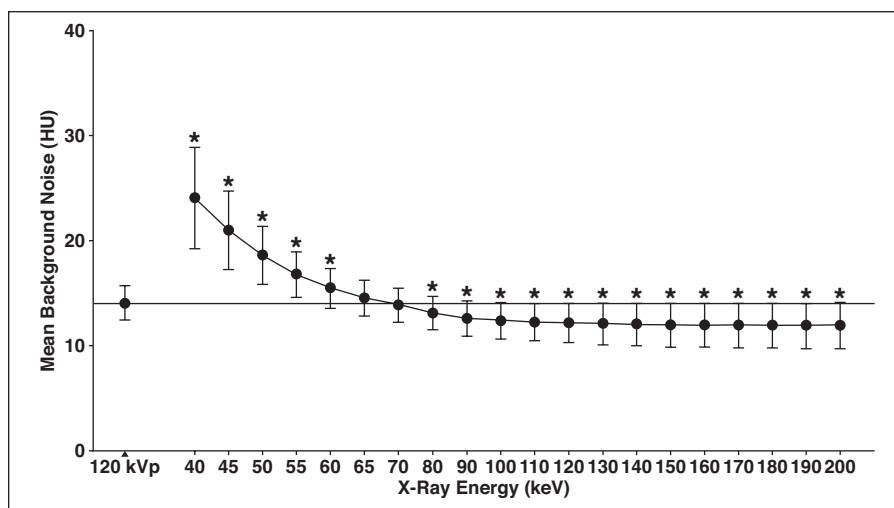


Fig. 3—Graph of mean background noise at each energy level in 79 patients with suboptimally enhanced pulmonary artery on chest CT. Mean noise levels at 65 and 70 keV were similar to those at 120 kVp. Compared with mean background noise at 120 kVp (horizontal line), mean noise level was significantly higher at 40–60 keV but was significantly lower at 80–200 keV (both, $p < 0.05$). For each energy level, asterisk indicates statistically significant difference compared with 120 kVp. ● = mean, whiskers = ± 1 SD.

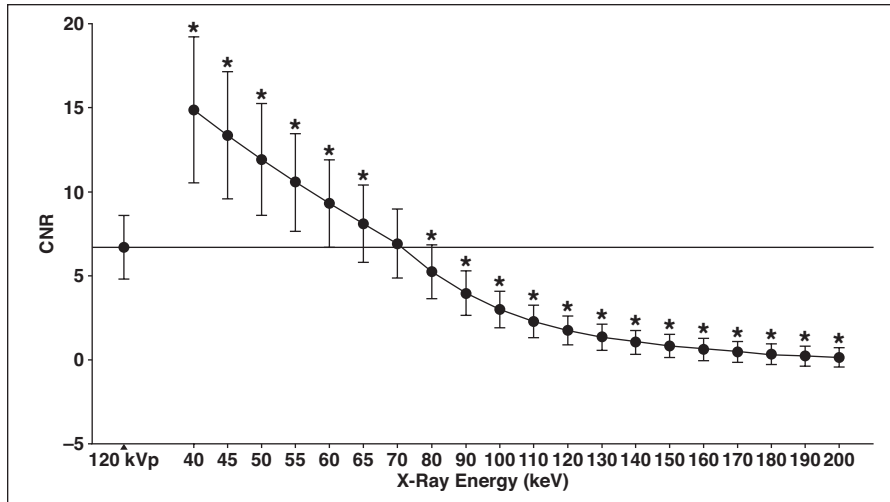


Fig. 4—Graph of mean contrast-to-noise ratio (CNR) of pulmonary artery at each energy level in 79 patients with suboptimally enhanced pulmonary artery on chest CT. Mean CNR at 40–65 keV was higher than that at 120 kVp ($p < 0.001$). However, mean CNR at 80–200 keV was significantly ($p < 0.001$) lower than mean CNR at 120 kVp (horizontal line). For each energy level, asterisk indicates statistically significant difference compared with 120 kVp. ● = mean, whiskers = ± 1 SD.

was significantly ($p < 0.001$) lower than that at 120 kVp (Fig. 4).

The mean SNR of the pulmonary artery on the 120-kVp image was 10.8 ± 2.0 . The mean SNR of the virtual monoenergetic images at 40–65 keV (12.1 ± 2.4 – 17.4 ± 4.5) was greater ($p < 0.001$) than that at 120 kVp (Fig. 5). VMI at 40 keV showed the highest CNR for all 79 patients and the highest SNR for 76 of 79 patients; the highest SNR was at 45 keV for the remaining three patients.

TABLE 2: Subjective Image Quality of 120-kVp and 40-keV Images of 79 Patients With Suboptimal Opacification of Pulmonary Artery on Routine Chest CT

Score ^a	120 kVp	40 keV	p^b
Image noise			0.23
0	4	14	
1	60	46	
2	15	19	
3	0	0	0.13
Image artifacts			
0	2	0	
1	56	55	
2	21	18	
3	0	6	

Note—Data are numbers of patients.

^aScore = 0, complete or almost complete absence of either noise or artifacts; score = 1, mild noise or artifacts; score = 2, moderate noise or artifacts; score = 3, marked noise or artifacts.

^bWilcoxon signed-rank test for comparison of image noise and artifacts between 120-kVp and 40-keV images.

Qualitative Image Analysis

Subjective image quality scores of the conventional 120-kVp image and the virtual monoenergetic images with the best CNR (i.e., 40-keV images) were not significantly different (Table 2).

In the six patients with PE, the reference standard detected PE in 12 segments among a total of 48 segments (Fig. 6). The diagnostic accuracy for detecting PE calculated on a per-segment basis was significantly higher for 40 keV (reader 1: AUC, 0.992; $p = 0.033$; reader 2: AUC, 0.986; $p = 0.043$) than that for 120 kVp (reader 1: AUC, 0.911; reader 2: AUC, 0.933) (Fig. 7 and Table 3). Interreader correlation was substantial ($\kappa = 0.75$ for 120 kVp, $\kappa = 0.79$ for 40 keV).

Discussion

Insufficient pulmonary vascular enhancement in chest CT can occur from various causes such as contrast extravasation, technical error, and patient-related factors. Despite

adequate adjustments of injection parameters, hemodynamic fluctuations of contrast enhancement during breathing can affect the attenuation of the pulmonary artery [11]. Transient interruption of contrast material is a flow-related artifact caused by increased venous return from the inferior vena cava during deep inspiration, leading to insufficient enhancement of the pulmonary artery in chest CT [1, 12, 13]. Transient interruption of contrast material was the most common cause of suboptimal opacification of the pulmonary artery in this study. The instruction to take a deep inspiration at the beginning of the examination might have caused transient interruption of contrast material in some of our patients.

For improved vascular attenuation, virtual monoenergetic images from DECT have been investigated [14]. However, traditional VMI algorithms are limited because of substantially increased noise and vulnerability to artifacts, especially at lower energy levels close to the k-edge of iodine at

TABLE 3: Sensitivity, Specificity, and AUC Values for Detecting Pulmonary Embolism (PE) in Six Patients With PE Confirmed by Pulmonary CT Angiography

Reader	Sensitivity (%) ^a	Specificity (%) ^a	AUC (95% CI)	p
Reader 1				0.033
120 kVp	75.0	94.4	0.911 (0.793–0.974) ^b	
40 keV	91.7	97.2	0.992 (0.911–1.000) ^b	
Reader 2				0.043
120 kVp	83.3	91.7	0.933 (0.822–0.985) ^b	
40 keV	91.7	94.4	0.986 (0.901–1.000) ^b	

^aFor these values, readings were dichotomized and scores of 1 and 2 were considered negative for PE, and scores of 3–5 were regarded as positive for PE.

^bDifference is statistically significant.

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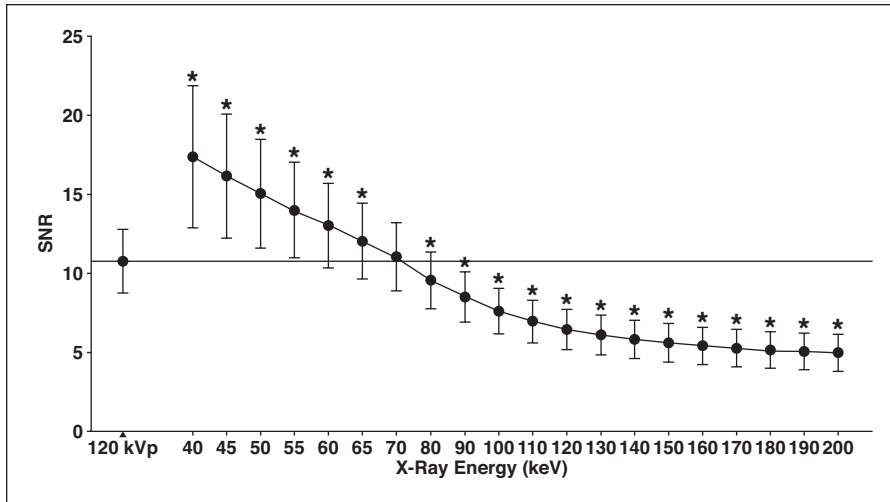


Fig. 5—Graph of mean signal-to-noise ratio (SNR) of pulmonary artery at each energy level in 79 patients with suboptimally enhanced pulmonary artery on chest CT. Mean SNR at 40–65 keV was significantly ($p < 0.001$) higher than mean SNR at 120 kVp (horizontal line). However, mean SNR at 80–200 keV was significantly lower than that at 120 kVp ($p < 0.001$). For each energy level, asterisk indicates statistically significant difference compared with 120 kVp. ● = mean, whiskers = ± 1 SD.

33 keV [15–17]. In pulmonary CTA with a reduced iodine load using a rapid tube voltage switching DECT technique, 50 keV has been chosen as the best level to balance vascular signal intensity and noise [4]. In a dual-source DECT system, the clinical utility of VMI at a very low energy level such as 40 keV has been limited until recent application of a noise optimization technique [18–20]. Our results showed a relatively low and consistent noise level over the entire energy spectrum in dual-layer detector spectral CT. The magnitude of the mean background noise even at 40 keV did not reach two times

that of the mean noise in a conventional 120-kVp image, which was much lower than the noise level measured on low-energy virtual monoenergetic images in DECT system using two tube voltage settings [16, 17]. This result may be because of two factors. The first factor is the effective iterative spectral reconstruction and noise removal algorithm in dual-layer detector spectral CT system. Simultaneously obtained, fully matched high- and low-energy projection data may facilitate noise suppression through anticorrelated noise removal in different datasets [21, 22]. The second factor is the energy overlap

between the high- and low-energy settings. The unavoidable overlap in spectral data by two detectors has been considered a major disadvantage of dual-layer detector CT [23, 24]. However, a larger overlap between high and low energies can contribute to noise reduction. A high-energy tail at the low-energy spectrum can keep the noise level low in low-energy images [24].

In the current study, VMI at 40 keV was the best in terms of CNR for all patients and SNR for most patients. The subjective image quality evaluated with regard to noise and artifacts between conventional 120-kVp and

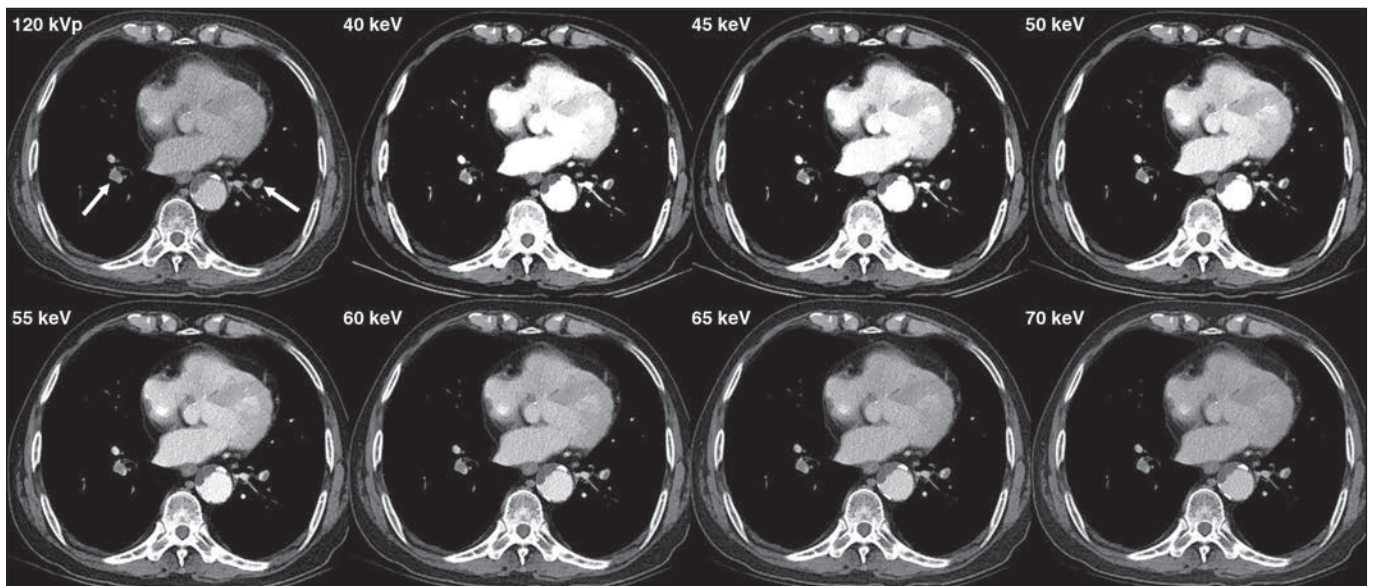


Fig. 6—72-year-old man who presented with nonspecific chest pain after drinking alcohol. During chest CT, insufficient amount of contrast agent was injected because of contrast extravasation and delayed scanning timing. Conventional 120-kVp image (top row, left image) shows suboptimal opacification of pulmonary artery (144 HU). There are suspicious emboli in both lower lobe segmental pulmonary arteries (arrows). However, differentiation between enhanced and nonenhanced portions of vessels is not clear. Virtual monoenergetic reconstruction was performed, and virtual monoenergetic images (all other images) show sufficient opacification of pulmonary artery (> 180 HU) was achieved at 40–60 keV. Pulmonary emboli are more clearly delineated on virtual monoenergetic images obtained at lower energy levels.

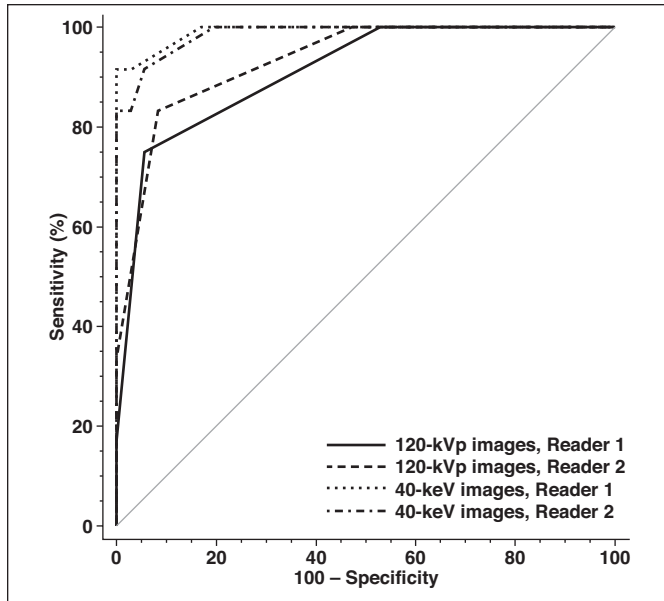


Fig. 7—Graph shows results of AUCs for 40-keV virtual monoenergetic images and 120-kVp CT images for detecting pulmonary embolism. Diagonal line is no discrimination line.

40-keV images was not different. Our observation corresponds with a recent study by Doerner et al. [25] showing that VMI at 40 keV from dual-layer detector spectral CT was the best among all image reconstructions of the chest including virtual monoenergetic images and a conventional 120-kVp image.

The capability of detecting PE even in nonangiographic chest CT is important because clinically unsuspected PE is frequently encountered in routine examinations. A missed PE diagnosis is a major cause of PE-related death [26–29]. To determine whether virtual monoenergetic images of good quality obtained from dual-layer detector spectral CT can yield additional diagnostic information, we compared the diagnostic performance for detecting PE of VMI at 40 keV and conventional 120-kVp images in patients with a suboptimally enhanced pulmonary artery. Our results revealed that 40-keV virtual monoenergetic images showed higher diagnostic accuracy in the detection of PE than conventional 120-kVp images. Recently, similar studies were performed using dual-source DECT [30–32]. Weiss et al. [30, 31] performed two studies evaluating image quality and diagnostic accuracy of the noise-optimized VMI technique in chest CT performed during the portal venous phase in oncology patients with PE. In their studies, CNR and SNR were the highest for the noise-optimized virtual monoenergetic images at 40 keV. However, reader's subjective confidence for detecting PE was the highest for the noise-optimized virtual monoener-

getic images at 55 keV. Leithner et al. [32] described the added value of the noise-optimized virtual monoenergetic images at 40 keV and perfusion DECT maps for the diagnosis of PE in patients with pulmonary CTA studies with suboptimal contrast enhancement. However, they did not perform individual analysis of the noise-optimized virtual monoenergetic images at 40 keV compared with the original polychromatic image.

In our study, CT scans were obtained with a routine protocol using 120 kVp without modifying the imaging protocol or patient selection process to obtain spectral information. We retrospectively collected chest CT examinations of patients with suboptimal opacification of the pulmonary artery, which occurred in various clinical situations. In dual-layer detector spectral CT, spectral data are captured during each CT examination. Virtual monoenergetic images can be extracted and extrapolated at wide energy levels, ranging from 40 to 200 keV, by simply clicking on the PACS integrated with the dedicated workstation. This simplified process for image acquisition and postprocessing in dual-layer detector spectral CT may widen practical uses of spectral information in daily practice such as for the evaluation of incidental findings or improvement of suboptimal studies.

This study has several limitations. The main limitation was that diagnostic performance of VMI at 40 keV for detecting PE was evaluated in only a small number of patients on a per-segment base. In addition, the diagnostic accuracy for detecting

PE in the segmental pulmonary artery was not assessed separately because PE was not the major clinically suspected entity in our study population, so the number of patients with PE confirmed by an additional pulmonary CTA examination was small. In addition, for practical reasons, we did not recommend additional pulmonary CTA in patients with no suspicion for PE according to the original CT or laboratory findings. To validate diagnostic performance and clinical relevance of virtual monoenergetic images from dual-layer detector spectral CT in detecting PE, further study with a larger number of patients in a prospective design should be performed. Second, in six patients with PE, not all pulmonary CTA studies were performed on the same day as the original chest CT examination. However, a standard reference for detecting PE was established by experienced chest radiologists using all available images. Third, the cut-off value of 180 HU for sufficient opacification of the pulmonary artery in chest CT was taken from previous studies, considering the highest possible attenuation of PE to be 180 HU [1, 33]. However, this value has not been validated in the detection of PE by clinical studies. Fourth, the conventional 120-kVp images were analyzed on a PACS, whereas the virtual monoenergetic images were analyzed on a dedicated workstation. However, we used the same slice thickness and tried to place ROIs on identical positions in both images by putting images side by side for each patient. Last, comparison of image qualities between 40-keV images and 120-kVp images could be biased because the evaluation was performed in consensus and the image features of two datasets were distinctly different. Although another study using the same scanner and reconstruction method showed similar results [25], further study is needed to compare image qualities between virtual monoenergetic images at different energy levels and conventional images obtained in dual-layer detector spectral CT.

In conclusion, virtual monoenergetic images obtained at low energy levels from dual-layer detector spectral CT can be used to obtain sufficient pulmonary artery opacification and can increase diagnostic performance for detecting PE in patients with a suboptimally enhanced pulmonary artery on chest CT. Low and consistent noise can be used to maintain image quality of virtual monoenergetic images at low energy levels with good CNR and SNR.

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References

1. Bae KT. Optimization of contrast enhancement in thoracic MDCT. *Radiol Clin North Am* 2010; 48:9–29
2. Ozawa Y, Hara M, Shibamoto Y. The frequency of insufficient contrast enhancement of the pulmonary artery in routine contrast-enhanced chest CT and its improvement with an increased injection rate: a prospective study. *J Thorac Imaging* 2011; 26:42–47
3. Delesalle MA, Pontana F, Duhamel A, et al. Spectral optimization of chest CT angiography with reduced iodine load: experience in 80 patients evaluated with dual-source, dual-energy CT. *Radiology* 2013; 267:256–266
4. Yuan R, Shuman WP, Earls JP, et al. Reduced iodine load at CT pulmonary angiography with dual-energy monochromatic imaging: comparison with standard CT pulmonary angiography—a prospective randomized trial. *Radiology* 2012; 262:290–297
5. Lee JW, Lee G, Lee G, et al. Effectiveness of adaptive statistical iterative reconstruction for 64-slice dual-energy computed tomography pulmonary angiography in patients with a reduced iodine load: comparison with standard computed tomography pulmonary angiography. *J Comput Assist Tomogr* 2016; 40:777–783
6. Riffel P, Haubenreisser H, Meyer M, et al. Carotid dual-energy CT angiography: evaluation of low keV calculated monoenergetic datasets by means of a frequency-split approach for noise reduction at low keV levels. *Eur J Radiol* 2016; 85:720–725
7. Beeres M, Trommer J, Frellesen C, et al. Evaluation of different keV-settings in dual-energy CT angiography of the aorta using advanced image-based virtual monoenergetic imaging. *Int J Cardiovasc Imaging* 2016; 32:137–144
8. Wichmann JL, Gillott MR, De Cecco CN, et al. Dual-energy computed tomography angiography of the lower extremity runoff: impact of noise-optimized virtual monochromatic imaging on image quality and diagnostic accuracy. *Invest Radiol* 2016; 51:139–146
9. Patino M, Prochowski A, Agrawal MD, et al. Material separation using dual-energy CT: current and emerging applications. *RadioGraphics* 2016; 36:1087–1105
10. Rassouli N, Chalian H, Rajiah P, Dhanantwari A, Landaras L. Assessment of 70-keV virtual monoenergetic spectral images in abdominal CT imaging: a comparison study to conventional polychromatic 120-kVp images. *Abdom Radiol (NY)* 2017; 42:2579–2586
11. Zhu J, Wang Z, Kim Y, et al. Analysis of contrast time-enhancement curves to optimise CT pulmonary angiography. *Clin Radiol* 2017; 72:340.e9–340.e16
12. Bernabé-García JM, García-Espasa C, Arenas-Jiménez J, Sánchez-Payá J, de la Hoz-Rosa J, Carreres-Polo JO. Has “respiratory coaching” before deep inspiration an impact on the incidence of transient contrast interruption during pulmonary CT angiography? *Insights Imaging* 2012; 3:505–511
13. Kay FU, Macedo AC, Chate RC, et al. Reduction of poor contrast enhancement of the pulmonary artery in computed tomography angiography using an alternative respiratory maneuver. *J Thorac Imaging* 2014; 29:107–112
14. Vlahos I, Chung R, Nair A, Morgan R. Dual-energy CT: vascular applications. *AJR* 2012; 199(suppl 5):S87–S97
15. Apfalter P, Sudarski S, Schneider D, et al. Value of monoenergetic low-kV dual energy CT datasets for improved image quality of CT pulmonary angiography. *Eur J Radiol* 2014; 83:322–328
16. Yu L, Leng S, McCollough CH. Dual-energy CT-based monochromatic imaging. *AJR* 2012; 199(suppl 5):S9–S15
17. Matsumoto K, Jinzaki M, Tanami Y, Ueno A, Yamada M, Kuribayashi S. Virtual monochromatic spectral imaging with fast kilovoltage switching: improved image quality as compared with that obtained with conventional 120-kVp CT. *Radiology* 2011; 259:257–262
18. Grant KL, Flohr TG, Krauss B, Sedlmair M, Thomas C, Schmidt B. Assessment of an advanced image-based technique to calculate virtual monoenergetic computed tomographic images from a dual-energy examination to improve contrast-to-noise ratio in examinations using iodinated contrast media. *Invest Radiol* 2014; 49:586–592
19. Meier A, Wurnig M, Desbiolles L, Leschka S, Frauenfelder T, Alkadhi H. Advanced virtual monoenergetic images: improving the contrast of dual-energy CT pulmonary angiography. *Clin Radiol* 2015; 70:1244–1251
20. Albrecht MH, Trommer J, Wichmann JL, et al. Comprehensive comparison of virtual monoenergetic and linearly blended reconstruction techniques in third-generation dual-source dual-energy computed tomography angiography of the thorax and abdomen. *Invest Radiol* 2016; 51:582–590
21. Alvarez RE, Macovski A. Energy-selective reconstructions in X-ray computerized tomography. *Phys Med Biol* 1976; 21:733–744
22. Ozguner O, Halliburton S, Dhanantwari A, Wen G, Utrup S, Jordan D. WE-FG-207B-11: objective image characterization of spectral CT with a dual-layer detector. *Med Phys* 2016; 43:3836
23. Lell MM, Wildberger JE, Alkadhi H, Damlakis J, Kachelriess M. Evolution in computed tomography: the battle for speed and dose. *Invest Radiol* 2015; 50:629–644
24. McCollough CH, Leng S, Yu L, Fletcher JG. Dual- and multi-energy CT: principles, technical approaches, and clinical applications. *Radiology* 2015; 276:637–653
25. Doerner J, Hauger M, Hickethier T, et al. Image quality evaluation of dual-layer spectral detector CT of the chest and comparison with conventional CT imaging. *Eur J Radiol* 2017; 93:52–58
26. Albrecht MH, Bickford MW, Nance JW Jr, et al. State-of-the-art pulmonary CT angiography for acute pulmonary embolism. *AJR* 2017; 208:495–504
27. Bach AG, Meyer HJ, Taute BM, Surov A. The frequency of incidental pulmonary embolism in different CT examinations. *Br J Radiol* 2016; 89:20150737
28. van Es N, Bleker SM, Di Nisio M. Cancer-associated unsuspected pulmonary embolism. *Thromb Res* 2014; 133(suppl 2):S172–S178
29. Chiu V, O’Connell C. Management of the incidental pulmonary embolism. *AJR* 2017; 208:485–488
30. Weiss J, Notohamiprodjo M, Bongers M, et al. Effect of noise-optimized monoenergetic postprocessing on diagnostic accuracy for detecting incidental pulmonary embolism in portal-venous phase dual-energy computed tomography. *Invest Radiol* 2017; 52:142–147
31. Weiss J, Notohamiprodjo M, Bongers M, et al. Noise-optimized monoenergetic post-processing improves visualization of incidental pulmonary embolism in cancer patients undergoing single-pass dual-energy computed tomography. *Radiol Med (Torino)* 2017; 122:280–287
32. Leithner D, Wichmann JL, Vogl TJ, et al. Virtual monoenergetic imaging and iodine perfusion maps improve diagnostic accuracy of dual-energy computed tomography pulmonary angiography with suboptimal contrast attenuation. *Invest Radiol* 2017; 52:659–665
33. Wittram C. How I do it: CT pulmonary angiography. *AJR* 2007; 188:1255–1261